

# TARGET-SPECIFIC RECOVERABILITY MAPS FOR REDUCED-LEAD ECG RECONSTRUCTION

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## ABSTRACT

Reconstructing missing electrocardiogram (ECG) leads from a reduced set is ill-posed, and a scalar reconstruction error says nothing about *which* named feature’s low-rank (dipolar) component, on *which* lead, is even recoverable. We give a *target-specific recoverability map*: for a waveform segment  $s$  (P/QRS/ST/T), an observed lead set  $S$ , and a target lead  $\ell$ , two closed-form numbers derived from one truncated SVD of the segment’s empirical rank-3 spatial (“dipolar”) sub-matrix—an *identifiability*  $\eta_{s,\ell}(S)$ , whose *exact* value (via the exact pseudoinverse) is zero iff the target’s dipolar component is recoverable from  $S$ , alongside a numerically  $\varrho$ -truncated variant  $\eta^\varrho$  that measures recoverability *within the retained rank-3 model*—and a *conditioning*  $\kappa_{s,\ell}(S)$  (its noise/observed-residual gain). The per-lead criterion is the classical estimability condition specialized to ECG. The robust findings: precordial ST is *exactly* non-identifiable from limb leads (the limb leads span the frontal plane, so  $\mathbf{M}_{s,S}$  has exact rank 2), and the graded normalized-identifiability ordering is bootstrap-stable for QRS. For the low-amplitude ST/T segments the fine ordering is *sensitive to the limb-lead weighting* (a duplicated-limb sensitivity analysis), so we report it with that caveat rather than as an exact graded claim. We attach distribution-free calibrated intervals for the empirically predictable residual (conformalized quantile regression, strict fold discipline, honestly stated within-group coverage), and observe that the total ST-threshold error rate’s point estimates range 51.1–53.7% across the evaluated reconstructors (an empirical observation, not a certified bound) while the false-positive/false-negative split is a reconstructor choice. The map turns “trust the reconstruction” into a per-feature, per-lead, calibrated statement.

**Index Terms**— electrocardiography, inverse problems, identifiability, conformal prediction, trustworthy machine learning

## 1. INTRODUCTION

Wearable and reduced-lead ECG acquisition makes lead *reconstruction* ubiquitous: recover the standard 12 leads from one, three, or six observed leads [1, 2]. A low aggregate error certifies nothing about *which* morphological features are

trustworthy: a cardiologist reads the P wave, the QRS complex, the ST segment and the T wave, each on specific leads.

We ask a sharper, reconstructor-independent question: *for a given observed lead set, which target lead’s dipolar morphology is even identifiable, and how well-conditioned is its recovery?* Each waveform segment’s instantaneous potential is approximately low rank across leads; we estimate a per-segment rank-3 spatial basis  $\mathbf{M}_s$  (top-3 singular vectors of the population lead covariance). We stress that  $\mathbf{M}_s$  is a *population-estimated (PCA)* object: on real PTB-XL its three principal angles to the classical vectorcardiographic (inverse-Dower) space are graded—one close (2–6°), one moderate (10–15°), and one substantially different (43–55°)—so we call it an empirical rank-3 subspace rather than a physical dipole (Sec. 4).

**Related work and contributions.** Classical work reconstructs missing leads (regression / VCG synthesis [3, 4, 5], deep and arbitrary-mask waveform-completion models [1, 2, 6, 7]) or *selects* optimal leads [8]; whether a given target is even recoverable is the classical estimability question [9]. Those systems produce a reconstruction; we instead certify, before any reconstruction, which target is recoverable and attach a graded, per-target-lead recoverability measure with calibrated per-record error. Our neural baseline is a *representative* masked-completion U-Net (same family as [1, 6]), not a state-of-the-art claim. We do not claim the underlying identity as new. Our contributions are: (1) a *graded*, per-target-lead recoverability map—continuous  $\eta_{s,\ell}(S)/\kappa_{s,\ell}(S)$  from a single truncated SVD that *rank* target leads (not a binary rank test), over an *empirically-estimated*  $\mathbf{M}_s$  that differs 43–55° from the textbook inverse-Dower dipole, with a truncation-tolerance sensitivity analysis and record-bootstrap CIs (Sec. 2); (2) a calibration layer attaching distribution-free intervals (a sound application of conformalized quantile regression + Mondrian [10], not a new method) to the map per feature and lead, with strict fold discipline and honest within-group (not per-diagnosis) coverage (Sec. 3); (3) a graded ST-safety study on PTB-XL—using the *exact* verdict that precordial ST is non-identifiable from limb leads (rank-2 frontal plane), with a total ST-threshold error rate whose point estimates range 51.1–53.7% across the evaluated reconstructors (an empirical observation, not a certified bound) and a false-positive/false-negative split that is a reconstructor

choice (Sec. 4).

## 2. THE RECOVERABILITY CERTIFICATE

**Model.** Within segment  $s$  write the population 12-lead vector as  $\mathbf{L}_s = \boldsymbol{\mu}_s + \mathbf{M}_s \mathbf{d} + \mathbf{r}_s$ , where  $\mathbf{M}_s \in \mathbb{R}^{12 \times 3}$  is the (population-estimated, PCA) dipolar basis,  $\mathbf{d}$  the dipole coordinates, and  $\mathbf{r}_s$  the non-dipolar residual. Observing  $S$  gives  $\mathbf{y}_S = \text{Sel}_S \mathbf{L}_s + \mathbf{n}$ , with  $\mathbf{M}_{s,S} = \text{Sel}_S \mathbf{M}_s$ . Write the exact Moore–Penrose pseudo-inverse  $\mathbf{M}_{s,S}^+$  and the exact row-space projector  $\mathbf{P}_{\text{obs}} = \mathbf{M}_{s,S}^+ \mathbf{M}_{s,S}$ . For numerical deployment we also use the *rank- $r$  truncation* at relative tolerance  $\varrho$ ,  $\mathbf{M}_{s,S}^{(\varrho)} = \mathbf{U}_r \boldsymbol{\Sigma}_r \mathbf{V}_r^\top$  keeping singular triplets with  $\sigma_i > \varrho \sigma_{\max}$ , with pseudo-inverse  $\mathbf{M}_{s,S}^{(\varrho)+}$  and projector  $\mathbf{P}_{\text{obs}}^{(\varrho)} = \mathbf{V}_r \mathbf{V}_r^\top$ .

**Proposition 1** (Exact per-target-lead dipolar identifiability and conditioning). *For each target lead  $\ell$  define  $\eta_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s (\mathbf{I} - \mathbf{P}_{\text{obs}})\|_2$  and  $\kappa_{s,\ell}(S) = \|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$  from the exact  $\mathbf{M}_{s,S}$ .*

(i) *Identifiability.*  $\eta_{s,\ell}(S) = 0$  iff  $\mathbf{e}_\ell^\top \mathbf{M}_s$  lies in the exact row space of  $\mathbf{M}_{s,S}$ ; then  $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$  is a deterministic function of  $\mathbf{M}_{s,S} \mathbf{d}$ —two dipoles with the same observation have the same lead- $\ell$  dipolar value, at any SNR. If  $\eta_{s,\ell}(S) > 0$  there is a  $\boldsymbol{\delta} \in \ker \mathbf{M}_{s,S}$  (an unobserved dipole direction) with  $\mathbf{e}_\ell^\top \mathbf{M}_s \boldsymbol{\delta} \neq 0$ ; then  $\mathbf{d}$  and  $\mathbf{d} + t\boldsymbol{\delta}$  produce identical observations for all  $t$  while lead  $\ell$ ’s dipolar value changes, so it is not identifiable at any SNR.

(ii) *Conditioning.* On the identifiable part, the lead- $\ell$  error from observation noise and observed non-dipolar residual obeys  $|\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{M}_{s,S}^+ (\text{Sel}_S \mathbf{r}_s + \mathbf{n})| \leq \kappa_{s,\ell}(S) (\|\text{Sel}_S \mathbf{r}_s\|_2 + \|\mathbf{n}\|_2)$ .

The global  $\kappa_s(S) = \|\mathbf{M}_s \mathbf{M}_{s,S}^+\|_2$  (spectral norm) upper-bounds every per-lead  $\kappa_{s,\ell}(S)$  (a row norm never exceeds the spectral norm; it is not their maximum).

*Exact vs.  $\varrho$ -identifiability (numerical deployment).* The deployed map computes  $\eta_{s,\ell}^{(\varrho)}, \kappa_{s,\ell}^{(\varrho)}$  from the truncation  $\mathbf{P}_{\text{obs}}^{(\varrho)}$ , so a singular direction observed only through a value below  $\varrho \sigma_{\max}$  is treated as unobserved. Then  $\eta_{s,\ell}^{(\varrho)} > 0$  means lead  $\ell$  is *not stably recoverable at resolution  $\varrho$*  (outside the retained numerical row space); it does *not* by itself imply the exact matrix has a null direction—a tiny-but-nonzero singular value is exactly identifiable ( $\eta = 0$ ) yet  $\varrho$ -unidentifiable ( $\eta^{(\varrho)} > 0$ ), as we verify numerically (`tests/test_exact_vs_rho.py`). We therefore call the  $\eta^{(\varrho)}$  heatmap the  *$\varrho$ -identifiability map*; for well-separated configurations the exact and  $\varrho$  verdicts coincide, and near-rank-deficient  $S$  are reported with a  $\varrho$ -sweep and record-bootstrap CIs. **The limb-6 flagship is exact:** the six limb leads are exact linear combinations of I and II

(Einthoven/Goldberger), so  $\mathbf{M}_{s,S}$  has exact rank 2 and precordial ST is *exactly* non-identifiable from limb leads at any SNR, independent of  $\varrho$ . This exactness is conditional on the a-priori rank-3 dipole model for  $\mathbf{M}_s$ : at retained rank 2 the limb-6 observation spans all of  $\mathbf{M}_s$  and the precordial verdict collapses to  $\eta=0$ ; the non-identifiability appears at rank  $\geq 3$  (the third dipolar direction), so “exact” means exact *given* rank 3.

*Relation to classical estimability.* Part (i) is the classical estimability criterion (Gauss–Markov / [9]) for the linear functional  $\mathbf{e}_\ell^\top \mathbf{M}_s \mathbf{d}$ ; we claim no novelty for this identity. The contribution (Sec. 4) is its use as a graded, empirically-estimated recoverability map with per-record calibrated error rates (Sec. 3).

**The non-dipolar residual (no independence claim on real data).** Let  $\mathbf{m}_s(\mathbf{y}_S) = \mathbb{E}[\mathbf{r}_s | \mathbf{y}_S]$  and  $\mathbf{u}_s = \mathbf{r}_s - \mathbf{m}_s(\mathbf{y}_S)$ , so  $\mathbb{E}[\mathbf{u}_s | \mathbf{y}_S] = \mathbf{0}$ . We do *not* assume  $\mathbf{u}_s \perp \mathbf{y}_S$ . Under squared loss the Bayes-optimal reconstruction error of the (vector) residual is

$$\inf_{\hat{\mathbf{r}}} \mathbb{E} \|\mathbf{r}_s - \hat{\mathbf{r}}(\mathbf{y}_S)\|_2^2 = \mathbb{E} [\text{tr Cov}(\mathbf{r}_s | \mathbf{y}_S)].$$

Part of  $\mathbf{r}_s$  (the *empirically predictable residual*) is recovered by a predictor trained on  $\mathbf{y}_S$  and metered with distribution-free calibrated intervals; the remainder is an *achievability gap*, established by a held-out achievability analysis for a stated predictor family—not declared unrecoverable a priori.

**Proposition 2** (Strict independence limit – synthetic model only). *If the data are generated so that a component  $\mathbf{u}$  is statistically independent of  $\mathbf{y}_S$  (an explicit synthetic construction), then  $p(\mathbf{u} | \mathbf{y}_S) = p(\mathbf{u})$ , the Bayes estimate of  $\mathbf{u}$  under squared loss is its prior mean, and  $\mathbb{E} \|\hat{\mathbf{u}} - \mathbf{u}\|_2^2 \geq \text{tr Cov}(\mathbf{u})$  for any estimator. This strict limit is invoked only for the synthetic model; on real ECG we make no such independence claim.*

## 3. DISTRIBUTION-FREE CALIBRATION

### 3.1. Calibrated intervals for the predictable residual

The prediction target is the *record-level segment-mean* off-dipole residual of the target lead (one scalar per record: the mean over the segment’s samples). We train a gradient-boosted quantile regressor of it on the observed leads’ segment means and apply Mondrian conformalized quantile regression [10] per  $(S, s, \ell)$  group, with strict fold discipline: basis and model on folds 1–7, the regressor’s `max_iter` selected by pinball loss on fold 8, conformal calibration on fold 9, and a single evaluation on fold 10 (nominal  $1-\alpha=0.90$ ). This is a sound *application* of established conformal machinery, not a new predictor; the contribution is attaching it, per feature and lead, to the recoverability map. Across the pre-registered  $(S, s, \ell)$  groups, fold-10 within-group coverage clusters near nominal (0.87–0.94; e.g. {I,II,V1,V3,V5}

QRS/V2 0.91, record-bootstrap CI [0.89, 0.93], mean width 0.150 mV).

**What is and is not guaranteed.** Mondrian CQR gives *within*-( $S, s, \ell$ )-group marginal coverage under exchangeability—not coverage conditional on diagnosis, which is not a conditioning variable. Diagnostic subgroups are therefore not guaranteed and the smallest ones fall below nominal: the worst is the HYP subgroup at coverage 0.72 ( $n=85$ ), the expected finite-sample behaviour when conditioning on a variable the calibration does not stratify on. We report worst-subgroup coverage rather than hide it under the aggregate band, and we make no distribution-free claim beyond exchangeability with the calibration fold.

## 4. EXPERIMENTS

**Data.** PTB-XL [11] (21,799 twelve-lead records, 100/500 Hz, 10-fold split) with P/QRS/ST/T delineation by NeuroKit2 [12].

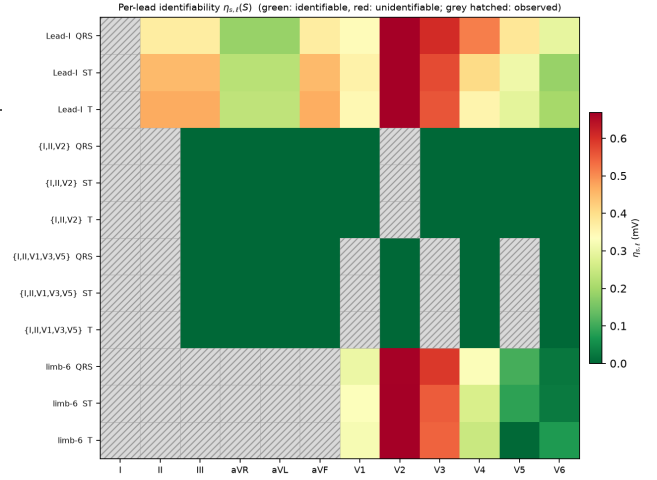
**Reproducibility.** Code, fixed and deterministic seeds, a pinned dependency manifest, result lineage (commit, a working-tree-clean flag, and dataset/split hashes in every JSON), and a one-command pipeline are released<sup>1</sup>; every number in the tables is regenerated from result JSON (no hand-typed values). The neural baseline is a 24→12-channel 1-D U-Net (base width 64, residual GroupNorm blocks, 60 epochs, Adam  $2 \times 10^{-4}$ , MSE on masked leads under random masks, 3 training seeds); the residual predictor is a gradient-boosted quantile regressor. The map is a deterministic function of the estimated  $\mu_s, M_s$  (record-level bootstrap CIs), not a held-out evaluation.

### 4.1. The recoverability map on PTB-XL

We estimate  $M_s$  (rank 3; the cardiac dipole is 3-D *a priori*, and the cumulative explained variance confirms rank 3 captures the bulk—a rank-sensitivity sweep  $r=2 \dots 5$  is reported in `results/recoverability_maps.json`) per segment on the training NORM records, with RECORD-LEVEL bootstrap CIs (resample records, refit  $M_s$ ;  $\varrho=10^{-2}$ ; Fig. 1). We plot the *normalized* identifiability  $\tilde{\eta}_{s,\ell} = \eta_{s,\ell} / \|e_\ell^\top M_s\|_2$ —a dimensionless relative unobserved geometric gain (a ratio of  $L_2$  norms, not a variance fraction). A single observed lead leaves every other lead unidentifiable; a dipole-spanning  $\{I, II, V2\}$  or  $\{I, II, V1, V3, V5\}$  makes all targets *exactly* identifiable ( $\eta=0$ ).

**The robust flagship is exact.** The six limb leads are exact linear combinations of I and II, so  $M_{s,S}$  for limb-6 has *exact* rank 2: precordial ST is *exactly* non-identifiable from limb leads at any SNR, independent of  $\varrho$  (Prop. 1).

<sup>1</sup>Anonymized for review; released on acceptance. Camera-ready artifacts are regenerated from a clean tree (the lineage `git_dirty` flag is asserted false).



**Fig. 1.** Recoverability map: normalized identifiability  $\tilde{\eta}_{s,\ell}(S)$  (left, dimensionless) and prior-conditional expected ambiguity in mV (right), colorblind-safe `cividis`; grey hatched = observed. For limb-6 all precordial leads are unidentifiable ( $\eta>0$ ); the fine ST/T ordering is weighting-sensitive (text).

**Graded ordering: stable for QRS, weighting-sensitive for ST/T.** A duplicated-limb sensitivity analysis (`results/lead_weighting`) refit on the 8 algebraically independent leads, on the *same* records) shows the V1–V6  $\tilde{\eta}$  ordering is bootstrap-stable for QRS (Spearman 1.00, CI lower bound 0.88; anterior-before-lateral in 0.98 of resamples) but *not* robust for the low-amplitude ST/T segments (ST Spearman point 0.89 with a wide bootstrap CI; anterior-before-lateral in only 0.77 of resamples). We therefore report the ST/T  $\tilde{\eta}$ /ambiguity magnitudes with this caveat and *do not* claim a fine ST/T ordering; the binary limb→precordial verdict above is the robust ST statement. The estimated subspace also depends on diagnosis (a NORM vs. MI principal-angle sensitivity is reported), so the map is a *NORM-trained reference*.

**Downstream ST cost across reconstructors.** We reconstruct the full 10-second waveform with three real continuous linear reconstructors (observed leads exact; ST at J+60 ms vs. the isoelectric PR-segment baseline [P offset–QRS onset], on fiducials from the observed Lead II shared between truth and reconstruction; median over beats; the PR-segment baseline fell back to the pre-QRS window in  $\approx 14\%$  of beats), and count ST-threshold events ( $|ST| \geq 0.1$  mV; false positive = crossing in reconstruction not truth, false negative = vice versa; Table 1). On 1498 common records the *total* wrong-event rate has point estimates ranging 51.1–53.7% across reconstructors and is *dominated by false negatives* (missed true crossings: limb→precordial recovery smooths ST toward the population), with the false-positive/false-negative split a reconstructor choice (paired record-bootstrap  $\Delta$  on the FP/FN split, and on the total, in `results/st_safety.json`). We do not derive a minimax/Bayes bound and make no equiv-

**Table 1.** Limb-6  $\rightarrow$  precordial (all V1–V6) ST reconstruction, 1498 common fold-10 records. FP/FN = false-positive/false-negative ST-threshold crossings (%); total = FP+FN. FN dominates (missed crossings); the split is a reconstructor choice. Auto-generated from `results/st_safety.json`.

| reconstructor | FP % | FN % | total % | ST  err (mV) |
|---------------|------|------|---------|--------------|
| dipolar       | 5.3  | 45.7 | 51.1    | 0.077        |
| ridge         | 1.7  | 52.1 | 53.7    | 0.076        |
| OLS           | 1.7  | 51.8 | 53.5    | 0.076        |

alence claim: we report the range of total-error point estimates, *not* a certified lower bound, and ST-threshold events, not diagnoses. Because the PR-segment baseline fell back to the pre-QRS window in  $\approx 14\%$  of beats, the exact ST magnitudes inherit that delineation limitation.

**Truncation-tolerance sensitivity.** Well-posed configurations are  $\varrho$ -stable ( $\{I, II, V1, V3, V5\}$  is rank 3 across  $\varrho \in \{10^{-4}, \dots, 10^{-1}\}$ ); near-rank -deficient sets are  $\varrho$ -sensitive and reported with the  $\varrho$ -sweep and bootstrap CIs. A tiny-but-nonzero singular value is exactly identifiable yet  $\varrho$ -unidentifiable, which is why we separate exact from  $\varrho$ -identifiability (Prop. 1).

## 4.2. Baselines and what the map predicts

Table 2 reports reconstruction RMSE (mV) of the unobserved target leads under a *single, identical protocol* for all methods: the same shared split (train folds 1–7, ridge  $\lambda$  selected on fold 8, test on NORM fold-10), the same 800 test records, per-timepoint waveform RMSE on the same segment indices, and *paired* record-bootstrap CIs on the step-to-step differences. We include a *representative* masked-completion baseline—an arbitrary-mask conditional 1-D U-Net trained with random lead masks (one model serves any configuration, no target-lead leakage), the same family as [1, 6, 7], run over three seeds (mean reported)—not a state-of-the-art claim.

On a dipole-spanning set the methods form a ladder in point estimates (the ridge  $\rightarrow$  U-Net step is within its paired CI)—prior-mean  $\rightarrow$  dipolar (Tier I)  $\rightarrow$  per-segment ridge (Tier I+II)  $\rightarrow$  U-Net—evidence of recoverable non-dipolar content (the calibrated Tier II layer), though the neural model’s margin over a simple ridge is small (QRS 0.164 vs. 0.167 mV on  $\{I, II, V1, V3, V5\}$ ; paired  $\Delta$  CI  $[-0.007, +0.000]$  mV). On **limb-6** the U-Net still *lowers* error over the prior mean (QRS 0.481  $\rightarrow$  0.238)—it recovers the predictable, largely population-correlated structure—but plateaus at  $1.46\times$  its spanning-set error (paired spanning  $\rightarrow$  limb-6  $\Delta$  CI  $[+0.070, +0.079]$  mV) and no reconstructor closes that gap. This residual gap is *consistent* with the unobserved dipolar coordinate the map’s  $\eta > 0$  marks as unidentifiable (we do not decompose the error into projected components, so we claim consistency, not equal-

**Table 2.** Reconstruction RMSE (mV) of target leads, QRS segment, under one identical per-timepoint protocol on NORM fold-10 (800 test records, shared split). “dipolar” = Tier I; “ridge” = per-segment ridge (Tier I+II); “U-Net” = representative masked-completion baseline (3-seed mean). Best in bold. On limb-6 even the neural baseline plateaus far above its identifiable-set error. Numbers auto-generated from `results/fair_baselines.json`.

| config                  | prior-mean | dipolar | ridge | U-Net        |
|-------------------------|------------|---------|-------|--------------|
| $\{I, II, V2\}$         | 0.467      | 0.236   | 0.220 | <b>0.183</b> |
| $\{I, II, V1, V3, V5\}$ | 0.481      | 0.183   | 0.167 | <b>0.164</b> |
| limb-6                  | 0.481      | 0.361   | 0.344 | <b>0.238</b> |

ity): capacity recovers the predictable part but not the missing coordinate, so a reconstruction that looked confident about anterior precordial morphology here would be inventing it. We therefore scope the identifiability claim to the dipolar component, not to a blanket “not recoverable.”

**Empirical, not physical, subspace.** On real PTB-XL the estimated  $M_s$  has three graded principal angles to the classical inverse-Dower vectorcardiographic space [3, 4]: one close ( $2\text{--}6^\circ$ ), one moderate ( $10\text{--}15^\circ$ ), and one substantially different ( $43\text{--}55^\circ$ , bootstrap CIs excluding small angles). We therefore treat  $M_s$  as an *empirical rank-3 spatial subspace* estimated from the data, not a physical cardiac dipole; all recoverability quantities are stated with respect to this estimated subspace. This empirical, data-driven subspace—rather than a fixed physiological transform—is what makes the map dataset-specific and, we verify, stable across PTB-XL and Chapman for the QRS segment (and robust to the duplicated-limb weighting).

## 5. CONCLUSION

Reduced-lead ECG reconstruction has a *known low-rank structure per segment*. Exploiting it gives a graded per-target-lead recoverability map—what is identifiable, how well-conditioned, and (via calibration) what interval the predictable residual admits—before and independent of any reconstructor. It states which lead configurations support recovery of a feature’s low-rank component, e.g. that precordial ST is exactly non-identifiable from limb leads, and reports the total ST-threshold error rate the evaluated reconstructors incur there (point estimates 51.1–53.7%, dominated by missed, not fabricated, crossings).

## 6. REFERENCES

- [1] Tomasz Gradowski and Teodor Buchner, “Deep learning model for ECG reconstruction reveals the information content of ECG leads,” *arXiv preprint arXiv:2502.00559*, 2025.

- [2] Gabriel V. Cardoso, Lisa Bedin, Josselin Duchateau, Rémi Dubois, and Eric Moulines, “Bayesian ECG reconstruction using denoising diffusion generative models,” *arXiv preprint arXiv:2401.05388*, 2024.
- [3] Lars Edenbrandt and Olle Pahlm, “Vectorcardiogram synthesized from a 12-lead ECG: superiority of the inverse Dower matrix,” *Journal of Electrocardiology*, vol. 21, no. 4, pp. 361–367, 1988.
- [4] Jan A. Kors, Gerard van Herpen, A. C. Sittig, and Jan H. van Bommel, “Reconstruction of the Frank vectorcardiogram from standard electrocardiographic leads: diagnostic comparison of different methods,” *European Heart Journal*, vol. 11, no. 12, pp. 1083–1092, 1990.
- [5] Stefan P. Nelwan, Jan A. Kors, Simon H. Meij, Jan H. van Bommel, and Maarten L. Simoons, “Reconstruction of the 12-lead electrocardiogram from reduced lead sets,” *Journal of Electrocardiology*, vol. 37, no. 1, pp. 11–18, 2004.
- [6] Alex Lence et al., “ECGrecover: A deep learning approach for electrocardiogram signal completion,” in *Proc. 31st ACM SIGKDD Conf. Knowledge Discovery and Data Mining (KDD)*, 2025, arXiv:2406.16901.
- [7] Xiaocheng Fang, Haoyu Wang, Jieyi Cai, Qinghao Zhao, Jun Li, Shenda Hong, et al., “ImputeECG: Deep learning reconstruction of complete 12-lead electrocardiograms from incomplete recordings for cardiac assessment,” *arXiv preprint arXiv:2607.05009*, 2026.
- [8] Siddharth Joshi and Stephen Boyd, “Sensor selection via convex optimization,” *IEEE Transactions on Signal Processing*, vol. 57, no. 2, pp. 451–462, 2009.
- [9] Henry Scheffé, *The Analysis of Variance*, Wiley, New York, 1959.
- [10] Yaniv Romano, Evan Patterson, and Emmanuel J. Candès, “Conformalized quantile regression,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, vol. 32.
- [11] Patrick Wagner, Nils Strodthoff, Ralf-Dieter Bousseljot, Dieter Kreiseler, Fatima I. Lunze, Wojciech Samek, and Tobias Schaeffter, “PTB-XL, a large publicly available electrocardiography dataset,” *Scientific Data*, vol. 7, no. 1, pp. 154, 2020.
- [12] Dominique Makowski, Tam Pham, Zen J. Lau, et al., “NeuroKit2: A Python toolbox for neurophysiological signal processing,” *Behavior Research Methods*, vol. 53, no. 4, pp. 1689–1696, 2021.